INCH-POUND

MIL-S-48747B (AR)
AMENDMENT 5
16 February 1995
SUPERSEDING
AMENDMENT 4
30 July 1986

MILITARY SPECIFICATION

SAFE AND ARMING DEVICE FOR MINES,
M718/M741 SYSTEM

This amendment forms a part of MIL-S-48747B (AR), dated 14 October 1982, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Beneficial Comments: Delete "DRDAR-QA" and substitute "AMSTA-AR-EDE-S".

PAGE 2

2.1.1, Add under SPECIFICATIONS; MILITARY:

"MIL-S-45743 - Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment

MIL-P-50884 - Printed Wiring, Flexible, General Specification for"

PAGE 4

3.10, Add new paragraph as follows:

"3.10 Flexible cable solder pads.

3.10.1 Lifted or overheated pads. The flexible cable shall contain no lifted, burned or overheated pads.

AMSC N/A FSC 1345

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PAGE 4 (Cont'd)

3.10.2 Solderability. When the flexible cable is tested for solderability in accordance with MIL-P-50884, the cable shall show full wetting and new solder coverage of at least 95% of conductor area not covered by insulating material or extruded adhesive. There shall be no distortion or delamination.

Solderability of each batch of flexible cable solder pads shall be demonstrated a maximum of thirty (30) days prior to assembly to S&A's.

3.10.3 Process control. All solder pad areas shall accept solder evenly, i.e., meet the solderability requirements of 3.10.2 when a solder iron from the assembly operation heats the pad area for a maximum of three (3) seconds, and solder from the assembly operation is applied during this period."

PAGE 6

TABLE II - Sheet 1 of 2: Delete and substitute new TABLE II, Sheet 1 of 3, included.

PAGE 7

TABLE II - Sheet 2 of 2: Delete and substitute new TABLE II, Sheet 2 of 3, included.

PAGE 7a

TABLE II: Add new Sheet 3 of 3 as page 7a, included.

PAGES 12 and 13

4.4.2.4: Delete in its entirety and substitute new paragraph included.

PAGES 16a and 16b

4.4.2.8: Add new paragraph included.

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4.4.3.5, Delete in its entirety and substitute new paragraph as follows:

"4.4.3.5 Solder workmanship. The contractor shall assure that the solder process/procedure at each solder assembly station does not contaminate or create a potential short circuit condition(s) on the bottom side of the flexible cable. The contractor shall examine the interface between the detonators and the cable, for: flux, excessive solder, or other unacceptable criteria specified in the solder standard listed on the drawing. The examination shall be performed on five (5) detonator/grounding clip/cable assemblies prior to the start of each shift, after every two (2) hours, and at the end of the shift until three (3) consecutive shifts have been accepted. Thereafter, the examination may be performed on two (2) assemblies prior to the start of each shift and one (1) sample after four (4) hours of production and at the end of the shift. Failure of the sample at any time will cause the examination to revert to the initial conditions until three (3) consecutive shifts have been accepted."

4.4.3.6, Add new paragraph as follows:

"4.4.3.6 Flexible cable solderability process control. The samples from 4.4.3.5 shall be used for this examination. Any solder pad failing to comply with the specified requirements shall have its flexible cable/detonator/grounding clip assembly classed defective."

4.4.3.7, Add new paragraph as follows:

"4.4.3.7 Detonator/clip assembly process control. The contractor shall maintain a positive process control to assure that both detonators and the grounding clip are fully seated in the S&A housing. For the purpose of this examination, fully seated is defined as having the two (2) detonators or the grounding clip flush to above flush by no more than .0025 inch while the other is fully seated. The samples from 4.4.3.5 shall be used to verify the process. Any assembly failing to comply with the specified requirement shall be classed defective."

4.4.3.7: Delete ".0025 inch" and substitute ".0035 inch".

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Delete 4.5.4 in its entirety and substitute the following:

"4.5.4 Radiographic examination. The safe and arming assemblies shall be radiographically examined (Ref. MIL-STD-453) using equipment which has image quality, such that it clearly defines all explosive components, and positively shows: the assembly and all components are in the fully non-armed conditions; the flexible cable is not pinched, twisted, or damaged; the MDF assembly has no evidence of damage.

Any assembly failing to comply with all specified requirements shall be classified defective, and removed from the lot. (Non-destructive test).

Radiographic equipment design submission shall be in accordance with paragraph 6.3."

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* 4.5.6.2, Add after sentence 3: "The test fixture, or the aft ejection charge assembly may be modified to allow for insertion of the assembly into the fixture. The modification shall be submitted to the Government for approval."

PAGE 30

* 4.5.8.5: Delete paragraph in its entirety.

PAGE 31

4.5.13, Delete in its entirety and substitute:

"4.5.13 Solder workmanship. The flexible cable/detonator/grounding clip assembly shall be examined under a stereo microscope at a four to ten power (4X to 10X) magnification. Lighting shall be as specified in MIL-S-45743 for inspection purposes. The assembly shall be examined for the specified solder workmanship requirements. Any assembly containing one (1) or more defects shall be classed defective. Any unit failing to comply with the soldering requirements at the start of the shift shall result in the contractor not proceeding with production until corrective action is taken and a subsequent sample passes. Any sample failing during the subsequent examinations or at the end of the shift shall result in all production following the last successful examination being rejected. (Destructive examination)."

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4.5.14, Add new paragraph as follows:

"4.5.14 Flexible cable solder pad process control and solderability. The solder pads to be soldered to the electronic lens assembly shall be tested for the specified requirement utilizing the soldering iron and solder used in the normal production assembly process. The pad shall demonstrate adherence to the solderability requirements during the maximum time period specified for heat application. Any pad failing to fully demonstrate full solderability shall be classed defective. Any unit failing to comply with the soldering requirements at the start of the shift shall result in the contractor not proceeding with production until corrective action is taken and a subsequent sample passes. Any sample failing during the subsequent examinations or at the end of the shift shall result in all production following the last successful examination being rejected. (Destructive examination)."

4.5.15, Add new paragraph as follows:

"4.5.15 Detonator/clip assembly process control. The contractor shall describe the process control procedures and gaging techniques required to assure compliance to the seating requirement. These procedures shall be forwarded to the Government for approval prior to use (see 6.3). Any assembly failing to comply with the requirement at the start of the shift shall result in the contractor not proceeding with production until corrective action is taken and a subsequent sample passes. Any sample failing during the subsequent examinations or at the end of the shift shall result in all production following the last successful examination being rejected. (Non-destructive examination)."

* 6.3: Delete "ARRADCOM, ATTN: DRSMC-QAT-I" and substitute "ARDEC, ATTN: AMSTA-AR-QAA-B."

* 6.5: Delete "ARRADCOM, ATTN: DRSMC-QAT-M" and substitute "ARDEC, ATTN: AMSTA-AR-QAA-B."

* 6.6: Delete "ARRADCOM, ATTN: DRSMC-QAT-M" and substitute "ARDEC, ATTN: AMSTA-AR-QAA-B."

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Safe and Arming Device for M718/M741 Systems	1 OF 3	See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Slider Assembly				
	(Dwg. 9278011)				
	Examination for defects	10		3.2	4.4.2.1
	Safe and Arming Subassembly				
	(Dwg. 9278015)				
	Examination for defects	10		3.2	4.4.2.2
	Non-arming test	10		3.2	4.5.1
	Non-arming test f/centrifugal weights	10		3.2	4.5.3
	Arming test	10		3.2	4.5.2
	Safe and Arming Assembly (Inprocess)				
	(Dwg. 9278015)				
	Examination for defects	10		3.2	4.4.2.4
	Closure firing pin stake push off test	5		3.2	4.5.7
	Closure slider stake push off test	5		3.2	4.5.7
	Solder workmanship under cable	25		3.2	4.5.13
	Flexible cable solder pad solderability	25		3.10.2	4.5.14
	Detonator/clip assembly seating	25		3.2	4.5.15

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Safe and Arming Device for M718/M741 Systems		SHEET 2 OF 3		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY: ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Safe and Arming Assembly (Dwg. 9278015)	10		3.2	4.4.2.8
	Examination for defects				
	Radiographic examination	10		3.3	4.5.4
	Jolt	5		3.4.1	4.5.8.1
	Jumble	5		3.4.2	4.5.8.2
	Transportation vibration	5		3.4.3	4.5.8.3
	Static detonator safety	15 (Note 1)		3.4.4	4.5.8.4
	Explosive tests	25 (note 2)		3.5.1	4.5.6.1
	Detonator resistance	25		3.2	4.5.5.1
	Detonator short circuit	25		3.2	4.5.5.2
	Chamber Assembly (Dwg. 92777986)				
	Examination for defects	10		3.2	4.4.2.5
	Five(5) each assemblies shall be tested at the temperature specified in MIL-STD-331. Note 2. 15 Ambient, 5 Hot, 5 Cold.				

NOTES:

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Safe and Arming Device for M718/M741 Systems	3 OF 3	See Below
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Aft Ejection Charge Assembly (Dwg. 9277858)				
	Examination for defects	25		3.2	4.4.2.6
	Functioning test	25		3.5.2	4.5.6.2
	Setback Pin Assembly (Dwg. 9317617)				
	Examination for defects	10		3.2	4.4.2.7
	Safe and Arming (Non-Hazardous) Assembly for Mines, M718/M741 System (Dwg. 9317620)				
	Examination for defects	108		3.2	MIL-S-63383
	Ballistic tests				
	Zone 3 -25°F	36 (4	rounds)	3.4.5	4.5.11
	Zone 8 70°F	36 (4	rounds)	3.4.5	4.5.11
	Zone 8 +145°F	36 (4	rounds)	3.4.5	4.5.11

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Safe and Arming Assembly (In-Process Examination)		SHEET 1 OF 2		DRAWING NUMBER: 9278015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Interrupter free to move		100%	3.2	Dwg. 9278015 (Note 4)
102	Chamber assembly improperly staked		100%	3.2	Visual
103	Grounding clip not fully seated			3.2	4.4.3.7/4.5.15
104	Detonators not fully seted			3.2	4.4.3.7/4.5.15
105	Solder workmanship (Note 1)		100%	3.2	Visual
106	Presence of all closure discs		100%	3.2	Visual/Manual
107	All main charge lead assemblies correctly inserted at staked		100%	3.2	Visual/Manual
108	Cable spacer improperly installed		100%	3.2	Visual
109	Closure firing pin stake push off test			3.2	4.4.3.2/4.5.7
110	Closure slider stake push off test			3.2	4.4.3.2/4.5.7
111	Closure, slider oriented improperly		0.40%	3.2	Visual
112	Detonator placement incorrect		100%	3.2	Visual

REMARKS: Solder workmanship of detonator terminals and grounding clip tab to flexible cable pads.

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TABLE II. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Safe and Arming Assembly (In-Process Examination)		SHEET 2 OF 2		DRAWING NUMBER: 9278015
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (Con'd)					
113	Solder workmanship under flexible cable improper			3.2	4.4.3.5/4.5.13
114	Flexible cable overheated or lifter solder pads	100%		3.10.1	Visual
115	Flexible cable solder pad solderability and process control			3.10.2/ 3.10.3	4.4.3.6/4.5.14
Minor					
201	Poor workmanship		1.00%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Safe and Arming Assembly	1 OF 2	9278015
			NEXT HIGHER ASSEMBLY: ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(Note 1)				
1	Assembly in partially armed, armed state or missing components		100%	3.3	4.5.4
2	Jolt	5/0/1		3.4.1	4.4.3.3/4.5.8.1
3	Jumble	5/0/1		3.4.2	4.4.3.3/4.5.8.2
4	Static detonator safety	5/0/1		3.4.4	4.4.3.3/4.5.8.4
Special					
A	"G" Weight lock spring missing or improperly positioned		100%	3.4.6	Visual
Major					
101	Shield assembly missing, loose, not staked or improperly installed		100%	3.2	Visual/Manual
102	Explosive test		100%	3.5.1	4.4.3.1.1/4.5.6.1
103	MDF Damaged		100%	3.2	Visual
104	Transportation vibration	5/0/1		3.4.3	4.4.3.3/4.5.8.3
105	Detonator resistance		100%	3.2	4.5.5.1
106	Detonator not shorted (Note 2)		100%	3.6	4.5.5.2
107	Foreign material on flexcable solder pads		100%	3.2	Visual
108	Cable spacer missing		100%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Safe and Arming Assembly				9278015
					NEXT HIGHER ASSEMBLY ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor 201	Poor workmanship		1.0%	3.9	Visual

NOTES: 1. Para. 4.5.12 applies whenever an inspection or test item is classed critical.
2. Short circuit shall be verified after resistance measurements have been taken and the short circuit has been replaced.

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6.8, Add new paragraph as follows:

"6.8 Special defect. A defect other than critical, that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or,

b. Prevent performance of the tactical function of a major end item."

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